

# AIX Box Software Architecture

## AIX Box Built-in Software System Architecture

### 1. System Overview

AIX Box is a decentralized edge node that integrates AI inference, distributed storage, hardware wallet, and advertising services. The software architecture consists of 6 layers from hardware to applications.

### 2. Layer Architecture

#### Application Layer

Components: AIX TV, Skill Store, TradeNet, Knowledge Commons, Firecrawl

#### AI Layer

Components: Pico Claw (inference), Hindsight (memory), xmemory (structured), MiniMind (training), Graphify (knowledge graph)

#### Agent Layer

Components: Hermes Agent (runtime), OpenOcta (optional), Bytebot (desktop), Turix-CUA (WeChat), TuriX (visual)

#### Financial Layer

Components: UTXO ledger, Coin Hour settlement, ATECC608B wallet, Oracle/anchoring

#### Network Layer

Components: Skywire (encrypted routing), EmerDNS (decentralized domain), BPFS/IPFS (storage), Yggdrasil (mesh)

#### System Layer

Components: Linux (Ubuntu/Debian), Docker/Podman, WASM sandbox, TEE/HSM

#### Hardware Layer

Components: RK3588/RK3588S chipset, ATECC608B security chip, SSD storage, WiFi/4G/Ethernet

# AIX Box Software Architecture

## 3. Core Components

### 3.1 Pico Claw - Local Inference Engine

- Model: Qwen2.5-7B (local deployment)
- Latency: 50-200ms
- Works offline without internet
- WASM sandbox execution
- Capability-based permission model

### 3.2 Coin Hour - Algorithmic Stablecoin

- Pegged to 1 RMB (non-redeemable to fiat)
- Auto-minted: 1 CH per hour per AIX unit held
- Transfer burn: 50% (25% to sender, 25% to receiver)
- Use cases: AI inference, village consumption, NFT trading
- Price stability: Arbitrage + TWAP + transaction limits + circuit breaker

### 3.3 UTXO Ledger - Consensus-less Accounting

- No consensus mechanism required
- Atomic transactions
- Local verification
- Hardware-level signing (ATECC608B)
- Immutable record of all transactions

### 3.4 Hermes Agent - Runtime Layer

- 24/7 operation
- Skill scheduling and execution
- Trajectory verification
- Integration with OpenClaw ecosystem
- Supports 200+ models via LiteLLM

# AIX Box Software Architecture

## 4. Economic Model - Coin Hour Mechanics

The Coin Hour economic model creates a closed-loop ecosystem:

Hold AIX → Auto-mint Coin Hour (24 CH/day per unit)

- Use for AI inference consumption
- Use for Nayej Village daily consumption
- Cannot cash out, circulates internally

Key Formula:  $X \times Y = 1$

- Coin Hour Price  $\times$  AIX Price = 1
- When AIX rises → Coin Hour cheaper → stimulates consumption
- When AIX falls → Coin Hour more valuable → holding incentive strengthens

## 5. Security Architecture

- **Hardware Level:**

ATECC608B physical private key storage

- **System Level:**

TEE (Trusted Execution Environment)

- **Application Level:**

WASM sandbox isolation

- **Network Level:**

Skywire end-to-end encryption

- **Economic Level:**

UTXO atomic verification

# AIX Box Software Architecture

## 6. Typical Usage Workflow

Example: Villager queries Nayej Village tourism information

Step 1: Voice input "Tell me about Nayej terraced fields"

↓

Step 2: Turix-CUA (voice recognition) → Hermes Agent (task scheduling)

↓

Step 3: Graphify (knowledge graph query) → Nayej Village tourism knowledge base

↓

Step 4: Pico Claw (local inference) → Generate response

↓

Step 5: xmemory (record query) → Update user preferences

↓

Step 6: UTXO ledger → Deduct 0.5 Coin Hour

↓

Step 7: Voice output response

Total Coin Hour Cost: 0.5 CH

Data Status: Never leaves the device

## 7. Comparison: Traditional vs AIX Box

| Dimension           | Traditional         | AIX Box                      |
|---------------------|---------------------|------------------------------|
| Data Sovereignty    | Cloud storage       | Local + BPFS distributed     |
| AI Inference        | Cloud API           | Pico Claw local (50-200ms)   |
| Economic Settlement | Fiat/Credit card    | Coin Hour closed loop        |
| Identity Auth       | Platform account    | ATECC608B hardware signature |
| Skill Extension     | Closed ecosystem    | Skill Store open market      |
| Network             | Centralized servers | Skywire P2P encrypted        |

Generated: 2026-05-05 | AIX Project | <https://aix2token.cloud>